

Robotics Assignments — Answer Key

NPTEL / SWAYAM • Weeks 1–8 • 80 questions with options and verified accepted answers. The correct option is highlighted in green.

Week 1 — Assignment 1

Introduction to Robotics: history, joints, manipulators & workspaces

1. Who is known as the father of robotics?

- (a) George Devol
- (b) ✓ Joseph Engelberger**
- (c) Isaac Asimov
- (d) Victor Scheinman

Accepted answer: (b)

2. "Sophia" (Humanoid Robot) was built by?

- (a) Honda
- (b) Odetics
- (c) ✓ Hanson Robotics**
- (d) NASA

Accepted answer: (c)

3. Robot is

- (a) a computer-controlled automatic machine used for piece production.
- (b) ✓ a computer-controlled automatic machine used for batch production.**
- (c) a computer-controlled automatic machine used for mass production.
- (d) none of the above.

Accepted answer: (b)

4. Ball and Socket Joint or Spherical Joint has

- (a) 1 degree of freedom (dof)
- (b) 2 dofs
- (c) ✓ 3 dofs**
- (d) 4 dofs

Accepted answer: (c)

5. Which one of the following statements is CORRECT?

- (a) Both Revolute Joint and Twisting Joint are linear in nature, and these are the same.
- (b) Both Revolute Joint and Twisting Joint are linear in nature, and these are different by their definitions.
- (c) Both Revolute Joint and Twisting Joint are rotary in nature, and these are the same.
- (d) ✓ Both Revolute Joint and Twisting Joint are rotary in nature, and these are different by their definitions.**

Accepted answer: (d)

6. By definition, an ideal Planar Manipulator has got

- (a) 1 dof
- (b) 2 dof
- (c) ✓ 3 dof**
- (d) 4 dof

Accepted answer: (c)

7. Which one of the following statements is TRUE?

(a) ✓ Degree of freedom (dof)/Mobility of a Serial Manipulator becomes equal to the summation of connectivity values of its joints.

(b) Degree of freedom (dof)/Mobility of a Parallel Manipulator becomes equal to the summation of connectivity values of its joints.

(c) Load carrying capacity of a Serial Manipulator is more than that of a Parallel Manipulator.

(d) Both the Serial Manipulator and Parallel Manipulator consisting of the same number of similar joints have exactly the same load carrying capacity.

Accepted answer: (a)

8. Spherical Coordinate or Polar Coordinate Robots have

(a) Two linear and one rotary movement

(b) ✓ One linear and two rotary movement

(c) Rotary movement about three independent axes

(d) Linear movement along three different axes

Accepted answer: (b)

9. The Workspace of a Spherical Coordinate Robot

(a) is a Cuboid.

(b) is a Cylindrical Annular Space.

(c) ✓ is the volume swept between two partial Spheres.

(d) None of the above

Accepted answer: (c)

10. Which one of the following statements is TRUE?

(a) Both Servo-controlled and Non-servo-controlled Robots use an Open-loop Control System.

(b) Both Servo-controlled and Non-servo-controlled Robots use a Closed-loop Control System.

(c) ✓ Servo-controlled Robot and Non-servo-controlled Robot have Closed-loop and Open-loop Control Systems, respectively.

(d) Servo-controlled Robot and Non-servo-controlled Robot have Open-loop and Closed-loop Control Systems, respectively.

Accepted answer: (c)

Week 2 — Assignment 2

Robot performance parameters, teaching, economics & grippers

1. Which one of the following statements is FALSE? Resolution of a Robot

(a) is similar to the least count of a measuring device.

(b) is the smallest allowable position increment of a robot.

(c) is a positive quantity only.

(d) ✓ may take the negative value also.

Accepted answer: (d)

2. Which one of the following statements is FALSE?

(a) ✓ Precision and accuracy of a robot are exactly the same.

(b) Precision and accuracy of a robot are not exactly the same.

(c) Repeatability of a robot is calculated based a previously taught point.

(d) Accuracy of a robot is determined with respect to a computed or target point.

Accepted answer: (a)

3. Telesurgery uses

- (a) One micro-capsule multi-legged robot.
- (b) ✓ Two Robots, one is called the Master and the other one is Slave.**
- (c) One free-flying robot.
- (d) None of the above.

Accepted answer: (b)

4. Off-line teaching method of a robot use

- (a) A robot simulator
- (b) A teach-pendant
- (c) ✓ A programming language**
- (d) A joystick

Accepted answer: (c)

5. To carry out Economic Analysis of a Robot, which one of the following statements is FALSE?

- (a) Depreciation of a Robot is defined as its falling value with age.
- (b) Pay-back period of a robot is preferred to be less than its Techno-economic Life.
- (c) Rate of Return on Investment is preferred to be more than the Rate of Bank Interest.
- (d) ✓ Rate of Return on Investment is preferred to be less than the Rate of Bank Interest.**

Accepted answer: (d)

6. Let us assume that Technical Life of a Robot is 30 years, and its Economic Life is 25 years. Its Techno-Economic Life will become equal to

- (a) ✓ 25 Years.**
- (b) $30 - 25 = 5$ Years
- (c) 30 Years
- (d) $30 + 25 = 55$ Years.

Accepted answer: (a)

7. A robotic gripper is called a passive gripper, if

- (a) It is not equipped with gripping jaws/fingers.
- (b) It is equipped with sensor(s).
- (c) ✓ It is not equipped with sensor(s).**
- (d) It is not equipped with an actuator.

Accepted answer: (c)

8. Remotely Operated Vehicle (ROV) and Autonomous Underwater Vehicle (AUV) used for the under-water applications, should have

- (a) Centralized Control only.
- (b) De-Centralized Control only.
- (c) ✓ Centralized and De-Centralized Controls, respectively.**
- (d) De-Centralized and Centralized Controls, respectively.

Accepted answer: (c)

9. In Remote Center Compliance (RCC) what should be the insertion angle?

- (a) ✓ Less than 45**
- (b) Greater than 45
- (c) Equal to 45
- (d) None of the above

Accepted answer: (a)

10. When designing a vacuum gripper, what material pairing is suggested for handling a hard object?

- (a) A magnetic material
- (b) ✓ A soft elastic material like rubber**
- (c) A hard material
- (d) None of the above

Accepted answer: (b)

Week 3 — Assignment 3

Transformations, rotation matrices & Denavit-Hartenberg notation

1. To represent orientation of one 3-D object, we use

- (a) ✓ 3x3 matrix.**
- (b) 4x4 matrix.
- (c) 5x5 matrix.
- (d) 6x6 matrix.

Accepted answer: (a)

2. Translation along Z axis by r units, that is, $\text{TRANS}(Z, r)$ is given by

- (a) Matrix with 1s on the diagonal and r in the 2nd row, 4th column.
- (b) ✓ Matrix with 1s on the diagonal and r in the 3rd row, 4th column.**
- (c) Matrix that is all zeros except r in the 2nd row 4th column and 1 in the bottom-right.
- (d) Matrix with 1s on the diagonal and r in the 1st row, 4th column.

Accepted answer: (b)

3. Orientation information of a 3-D object can be represented in 3-D space in

- (a) Roll-Pitch-Yaw System.
- (b) Euler Angles.
- (c) Cartesian Coordinate System.
- (d) ✓ all of the above.**

Accepted answer: (d)

4. A 4x4 matrix representing rotation about Y axis by an angle theta, and translation along the X, Y and Z axes by l, m and n units, respectively, is given by

- (a) ✓ Option a (rotation about Y with cos on rows 1 & 3, +Sin in row 1 / -Sin in row 3, translations l, m, n).**
- (b) Option b
- (c) Option c
- (d) Option d

Accepted answer: (a)

5. Which one of the following statements is FALSE?

- (a) ✓ TRANS (Translation) operators are not commutative in nature.**
- (b) Rotation matrix can be represented using Euler angles.
- (c) Position information can be represented in spherical coordinate system.
- (d) Rotation matrix can be represented using the concept of Roll, Pitch and Yaw angles.

Accepted answer: (a)

6. In Denavit-Hartenberg Notations, Link and Joint parameters are used to represent

- (a) the structure of a Link.
- (b) the relative positions of neighboring links.
- (c) ✓ the structure of a Link and relative positions of neighboring links, respectively.**
- (d) the relative positions of neighboring links and structure of a link, respectively.

Accepted answer: (c)

7. In robot kinematics, the concept of a 'Screw' is used to describe a combined translation and rotation motion. Which of the following represents the correct notation for 'Screw'?

- (a) ✓ ROT(Z, theta) TRANS(Z, d)**
- (b) TRANS(Z, d) ROT(Z, theta)
- (c) TRANS(Z, d) ROT(Z, theta) TRANS(Z, d)
- (d) ROT(Z, theta) TRANS(Z, d) ROT(Z, theta)

Accepted answer: (a)

8. In Denavit-Hartenberg notations, the Joint Angle

- (a) can have positive values only.
- (b) can have negative values only.
- (c) can have zero value only.
- (d) ✓ can have either positive value or negative value or zero value.**

Accepted answer: (d)

9. Consider a 2-D rotation matrix R (with Cos and Sin of theta). Which of the following statements is true about the inverse of the rotation matrix, denoted as R^{-1} ?

- (a) ✓ R^{-1} is equal to the transpose of R .**
- (b) R^{-1} is equal to R itself.
- (c) R^{-1} is equal to negative of R .
- (d) R^{-1} is equal to identity matrix.

Accepted answer: (a)

10. The concept of Roll, Pitch and Yaw angles is used to represent the rotation of frame {B} w.r.t. reference frame {U}, expressed by the given 3x3 rotation matrix. The angles of rolling (alpha), pitching (beta) and yawing (gamma) are found to be

- (a) ✓ alpha = 0 deg; beta = 0 deg; gamma = 60 deg**
- (b) alpha = 45 deg; beta = 60 deg; gamma = 45 deg
- (c) alpha = 0 deg; beta = 30 deg; gamma = 45 deg
- (d) alpha = 60 deg; beta = 0 deg; gamma = 0 deg

Accepted answer: (a)

Week 4 — Assignment 4

Trajectory planning, forward/inverse kinematics, singularity & Jacobian

1. Trajectory Planning of a Robot

- (a) aims to determine its collision-free path, while moving from an initial position to a final position.
- (b) ✓ aims to determine the time histories of position, velocity and acceleration of the Joints for ensuring their smooth variations.**
- (c) is not required to carry out at all.
- (d) is to be done before carrying out its Kinematic analysis.

Accepted answer: (b)

2. Forward Kinematics of a Serial Manipulator

(a) aims to determine its Joint Angle(s) and/or Link-Offsets for the given Position and Orientation information.

(b) aims to estimate its required Joint Force(s) and/or Torque(s).

(c) ✓ aims to determine the Position and Orientation of its End-effector with respect to the Base Coordinate System.

(d) aims to determine its Joint Angle(s) and/or Link-Offsets for the known Joint Force(s) and/or Torque(s) information.

Accepted answer: (c)

3. Which one of the following statements is FALSE?

(a) Trajectory planning problem is different from robot motion planning problem.

(b) ✓ Robot's trajectory planning problem and its motion planning problem are the same.

(c) Trajectory planning problem aims to ensure smooth variations at different robotic joints.

(d) Robot's motion planning problem aims to determine its collision-free path among the obstacles.

Accepted answer: (b)

4. In the field of robotics, what methods can be employed to address a robot's trajectory planning problem?

(a) Only utilizing the Cartesian coordinate system.

(b) Solely relying on the Joint space scheme.

(c) ✓ Utilizing both the Cartesian coordinate system and Joint space scheme.

(d) Completely disregarding both the Cartesian coordinate system and Joint space schemes.

Accepted answer: (c)

5. To fit a fifth-order polynomial function for a Robotic Joint, we need to have at least

(a) ✓ six known conditions.

(b) five known conditions.

(c) four known conditions.

(d) three known conditions.

Accepted answer: (a)

6. Which one of the following statements is FALSE?

(a) ✓ Pure Linear Trajectory function is preferred to that with the parabolic blends at its two ends.

(b) Linear Trajectory function with the parabolic blends at its two ends is preferred to the pure Linear Trajectory function.

(c) Trajectory function may be a nonlinear function other than Polynomial.

(d) The tip of a Robotic link will have nonlinear movement, even if its rotary joint follows a linear trajectory function.

Accepted answer: (a)

7. What is the purpose of studying the singularity condition of a manipulator?

(a) It is always advantageous to achieve singularity for its wide applications.

(b) The main objective is to increase the degrees of freedom (DOF) of the manipulator.

(c) ✓ The aim is to identify and analyze points in the manipulator's workspace, where it loses degrees of freedom.

(d) The concept of Jacobian matrix cannot be used to determine the singularity condition.

Accepted answer: (c)

8. For an industrial serial manipulator with 6 degrees of freedom (dof), what can be said about its forward kinematics and inverse kinematics?

(a) ✓ The forward kinematics problem has a unique solution, while the inverse kinematics problem has multiple solutions.

(b) The forward kinematics problem has multiple solutions, while the inverse kinematics problem has a unique solution.

(c) Both the forward kinematics problem and the inverse kinematics problem have a unique solution.

(d) Both the forward kinematics problem and the inverse kinematics problem have multiple solutions.

Accepted answer: (a)

9. Inverse Kinematics problem of MINIMOVER will have

(a) one solution only.

(b) two solutions.

(c) no solution.

(d) ✓ multiple solutions.

Accepted answer: (d)

10. Which one of the following statements is FALSE?

(a) Jacobian Matrix contains the information related to multi-dimensional form of the derivatives.

(b) To ensure non-singular configuration of a Manipulator, its Jacobian Matrix must be invertible.

(c) ✓ To ensure non-singular configuration of a Manipulator, its Jacobian Matrix need not be invertible.

(d) Jacobian Matrix of a Serial Manipulator can be used for checking its Singularity condition and Control, as well.

Accepted answer: (c)

Week 5 — Assignment 5

Robot dynamics: Lagrange-Euler, Coriolis force, moment & product of inertia

1. Forward dynamics is a method used to:

(a) Calculate the forces and torques acting on a system given its kinematics.

(b) ✓ Calculate the displacement, velocities and accelerations of a system given the applied forces and torques.

(c) Determine the equilibrium conditions of a system.

(d) Calculate the mass distribution of a system.

Accepted answer: (b)

2. According to Lagrange-Euler formulation, force (F) acting on a linear robotic joint can be determined as follows: (L = Lagrangian, d = joint displacement, d-dot = its time derivative)

(a) $F = \frac{d}{dt} \left(\frac{dL}{dd} \right) - \frac{dL}{d(d\text{-dot})}$

(b) $F = \frac{d}{dt} \left(\frac{dL}{dd} \right) + \frac{dL}{d(d\text{-dot})}$

(c) ✓ $F = \frac{d}{dt} \left(\frac{dL}{d(d\text{-dot})} \right) - \frac{dL}{dd}$

(d) $F = \frac{d}{dt} \left(\frac{dL}{d(d\text{-dot})} \right) + \frac{dL}{dd}$

Accepted answer: (c)

3. Under what circumstances does the Coriolis force occur?

(a) ✓ When there is a sliding component on a rotary link.

(b) When there is rotational motion without any sliding component.

(c) When there is a linear motion on a linear link.

(d) When there is a bending component on a rotary link.

Accepted answer: (a)

4. Why is dynamic analysis of a robot important?

- (a) To ensure the robot follows a path without collisions.
- (b) ✓ To understand the force and/or torque history of a robotic joint over time.**
- (c) To achieve smooth and jerk-free movement of the robot's joints.
- (d) To study the positions of different robotic links, without considering the underlying movement reasons.

Accepted answer: (b)

5. Moment of inertia can have

- (a) ✓ positive value only.**
- (b) negative value only.
- (c) zero value only.
- (d) either positive or negative or zero value.

Accepted answer: (a)

6. Fill in the blank. _____ theorem is used to calculate the moment of inertia of a link about an axis located at the center of its mass.

- (a) ✓ Parallel axis**
- (b) Perpendicular axis
- (c) Tangent axis
- (d) Multi-axis.

Accepted answer: (a)

7. For a robotic link of length l , mass m and circular cross-section of radius r , the 4x4 inertia tensor can be determined as follows:

- (a) Option a (diagonal terms with $m \cdot r^2/2$, off-diagonal $-m \cdot l/2$, m in last position).
- (b) Option b
- (c) ✓ Option c (correct inertia tensor)**
- (d) Option d

Accepted answer: (c)

8. Product of inertia (P_{xy}) of a given rectangular shape (height a , base b) with respect to X and Y axis is calculated as

- (a) $P_{xy} = a^2 b^2 / 2$
- (b) ✓ $P_{xy} = a^2 b^2 / 4$**
- (c) $P_{xy} = a^2 b^2 / 8$
- (d) $P_{xy} = a^2 b^2 / 16$

Accepted answer: (b)

9. Consider a 2-dof serial manipulator with two rotary joints. The joint angles of the first and second rotary joints are denoted by θ_1 and θ_2 . Torque for the first joint τ_1 depends on

- (a) θ_1 only.
- (b) $(\theta_1 + \theta_2)$ only.
- (c) θ_1 and $(\dot{\theta}_1)$ only.
- (d) ✓ θ_1, θ_2 , their velocities and their accelerations (all six terms).**

Accepted answer: (d)

10. Fig. A shows a 1-DoF manipulator arm of circular cross-section (diameter d , length L , mass m at its mass-center, g = acceleration due to gravity). Its joint torque is determined as follows:

- (a) $(1/16) m L^2 (\ddot{\theta}) + (1/2) m d^2 (\ddot{\theta}) + (1/2) m g L \cos(\theta)$
- (b) ✓ $(1/3) m L^2 (\ddot{\theta}) + (1/16) m d^2 (\ddot{\theta}) + (1/2) m g L \cos(\theta)$**
- (c) $(1/3) m L^2 (\ddot{\theta}) + (1/16) m d^2 (\ddot{\theta}) + m g L \cos(\theta)$
- (d) $(1/16) m L^2 (\ddot{\theta}) + (1/2) m b^2 (\ddot{\theta}) + m g L \cos(\theta)$

Accepted answer: (b)

Week 6 — Assignment 6

Robot sensors: encoders, proximity, force/moment sensors & calibration

1. Which one of the following statements is NOT accurate regarding optical encoders?

- (a) Incremental optical encoder uses two photodetectors.
- (b) Resolution of absolute optical encoder depends on the number of concentric rings used.
- (c) In absolute optical encoder, the number of photo-detector is kept equal to that of concentric rings.
- (d) ✓ Resolution of incremental optical encoder depends on the number of photodetectors.**

Accepted answer: (d)

2. In engineering applications, a Micro-switch is most accurately classified as which type of sensor?

- (a) acceleration sensor
- (b) position sensor
- (c) velocity sensor
- (d) ✓ touch sensor**

Accepted answer: (d)

3. Which one of the following sensors is classified as a proximity sensor?

- (a) ✓ capacitive sensor**
- (b) Linear variable differential transformer (LVDT)
- (c) Micro switch
- (d) Limit switch

Accepted answer: (a)

4. Which sensor is suitable for both ferro-magnetic and non-ferro-magnetic objects?

- (a) Inductive sensor.
- (b) Hall effect sensor.
- (c) ✓ Capacitive sensor.**
- (d) None of the above.

Accepted answer: (c)

5. For an absolute optical encoder with 'n' concentric rings, the resolution becomes equal to

- (a) $(1 / n^2) \times 360$ deg
- (b) ✓ $(1 / 2^n) \times 360$ deg**
- (c) $(1 / n^4) \times 360$ deg
- (d) $(1 / n) \times 360$ deg

Accepted answer: (b)

6. Which one of the following statements is FALSE regarding mechanical type of force/moment sensor?

- (a) It consists of some strain gauges mounted on four deflection bars.
- (b) It should be operated within the elastic limit of the material of its deflection bars.
- (c) ✓ It does not require any calibration.**
- (d) It uses potentiometer to measure the deflection of deflection bar.

Accepted answer: (c)

7. What is the term used to describe the process of correlating a measured quantity with a known physical parameter?

- (a) Comparison
- (b) Inspection
- (c) Analysis

(d) ✓ Calibration

Accepted answer: (d)

8. Identify the internal sensor from the options given below.

- (a) acoustic sensor
- (b) ✓ velocity sensor**
- (c) visual sensor
- (d) proximity sensor

Accepted answer: (b)

9. In which type of sensor is the triangulation method commonly utilized?

- (a) ✓ range sensor**
- (b) proximity sensor
- (c) potentiometer
- (d) force/moment sensor

Accepted answer: (a)

10. What type of control system is typically utilized to implement the Partitioned Control Scheme for motor control?

- (a) Opened-loop Control System.
- (b) ✓ Closed-loop Control System.**
- (c) Both Opened-loop Control System as well as Closed-loop Control System.
- (d) Neither Opened-loop Control System nor Closed-loop Control System.

Accepted answer: (b)

Week 7 — Assignment 7

Robot / computer vision, image processing & path-finding methods

1. What is the purpose of robot vision or computer vision?

- (a) To control the robot's movement
- (b) To enhance the robot's speech recognition
- (c) ✓ To collect information about the environment**
- (d) To improve the robot's decision-making capabilities

Accepted answer: (c)

2. What is frame grabbing in computer vision?

- (a) ✓ Image stored in computer as an array of picture elements or pixels.**
- (b) Extracting features from an image
- (c) Applying filters to an image
- (d) Converting images to grayscale

Accepted answer: (a)

3. In computer vision, Laplacian operator is utilized for

- (a) Frame grabbing
- (b) Thresholding
- (c) ✓ Edge detection**
- (d) Preprocessing

Accepted answer: (c)

4. Which one of the following operations/methods is not related to robot vision?

- (a) Frame grabbing
- (b) Segmentation
- (c) pre-processing
- (d) ✓ Cell decomposition**

Accepted answer: (d)

5. Which one of the following sequences is correct in digital image processing?

- (a) Image acquisition - Thresholding - Frame grabbing - Pre-processing - Digitizing - Edge detection
- (b) Image acquisition - Digitizing - Edge detection - Pre-processing - Thresholding - Frame grabbing
- (c) ✓ Image acquisition - Digitizing - Frame grabbing - Pre-processing - Thresholding - Edge detection**
- (d) Image acquisition - Pre-processing - Frame grabbing - Digitizing - Thresholding - Edge detection

Accepted answer: (c)

6. Behavior-based Robotics uses the principle of

- (a) ✓ Reactive Control Strategy.**
- (b) Potential Field Approach.
- (c) Path Velocity Decomposition.
- (d) Relative Velocity Approach.

Accepted answer: (a)

7. An image is given in digital form by light-intensity values at different pixels. The pre-processed value for the second-row, third-column element (82), obtained using a 3x3 Laplacian mask $\begin{bmatrix} -1 & -1 & -1 \\ -1 & +8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$, is found to be equal to

- (a) 57
- (b) 48
- (c) ✓ 59**
- (d) 85

Accepted answer: (c)

8. An image is given in digital form by light-intensity values at different pixels. The pre-processed value for the second-row, third-column element (74), obtained using median filtering with a 3x3 neighborhood, is found to be equal to

- (a) 68
- (b) ✓ 56**
- (c) 84
- (d) 49

Accepted answer: (b)

9. Which one of the following methods used for solving the find-path problem in robotics utilizes the concept of configuration space approach?

- (a) Tangent graph
- (b) Visibility graph
- (c) ✓ Cell decomposition**
- (d) Voronoi diagram

Accepted answer: (c)

10. Fine motion planning problem is known as

- (a) Gross motion planning problem.
- (b) ✓ Compliant motion planning problem.**
- (c) Free space motion planning problem.
- (d) Trajectory planning problem.

Accepted answer: (b)

Week 8 — Assignment 8

Motion planning, intelligent robots, biped walking & compactness

1. What is the primary application of the Voronoi diagram in robotics?

- (a) ✓ Solving find-path problems**
- (b) Solving motion-planning problems
- (c) Solving dynamic motion planning problems
- (d) Solving analytical problems in robotics

Accepted answer: (a)

2. An intelligent robot should have

- (a) adaptive controller only.
- (b) adaptive motion planner only.
- (c) adaptive motion planner but not adaptive controller.
- (d) ✓ Both adaptive motion planner and adaptive controller.**

Accepted answer: (d)

3. Which graph-based technique is commonly used for solving dynamic motion planning problems in robotics?

- (a) Voronoi Diagram
- (b) Visibility Graph
- (c) ✓ Accessibility graph**
- (d) Tangent Graph

Accepted answer: (c)

4. In biped walking, the concept of Zero-Moment Point (ZMP) is used to

- (a) determine its static stability margin.
- (b) ✓ determine its dynamic stability margin.**
- (c) carry out its joint torque calculation.
- (d) carry out its kinematic analysis.

Accepted answer: (b)

5. In the field of robotics, the recommended order of studying fundamental concepts (1. Trajectory Planning, 2. Kinematics, 3. Dynamics, 4. Control Scheme) is

- (a) 1 -> 2 -> 3 -> 4
- (b) ✓ 2 -> 1 -> 3 -> 4**
- (c) 3 -> 1 -> 2 -> 4
- (d) 3 -> 2 -> 4 -> 1

Accepted answer: (b)

6. How is artificial intelligence incorporated in robots?

- (a) ✓ incorporated in the form of software and implemented using hardware.**
- (b) incorporated in the form of hardware and implemented using software.
- (c) incorporated and implemented using neither the software nor hardware.
- (d) incorporated and implemented through a teaching method.

Accepted answer: (a)

7. What is the primary requirement for a humanoid robot standing at a particular location?

- (a) It is to be kinematically stable only.
- (b) It does not need power at its various joints.
- (c) It need not be either kinematically or dynamically stable.

(d) ✓ It is to be dynamically stable.

Accepted answer: (d)

8. Which one of the following statements is FALSE?

- (a) A biped robot consumes more energy while ascending a staircase compared to its descending the same.
- (b) ✓ A biped robot consumes less energy while ascending a staircase compared to its descending the same.**
- (c) Ascending a staircase by a biped robot will be the easier task in terms of its stability (balance) compared to its descending.
- (d) Both the power consumption and stability margin are the important responses to be calculated for the biped walking.

Accepted answer: (b)

9. Which type of robot is recommended for its movement on a terrain that is neither very smooth nor very rough?

- (a) ✓ Tracked vehicle type of Robot**
- (b) Six-legged Robot
- (c) Four-legged Robot
- (d) Wheeled Robot

Accepted answer: (a)

10. What are the compactness values of the circular (Radius = 6 units) and rectangular (Breadth = 3 units, Length = 5 units) objects, respectively?

- (a) 10.06 and 15.56
- (b) 13.07 and 14.06
- (c) ✓ 12.57 and 17.07**
- (d) 15.06 and 13.07

Accepted answer: (c)